Memory Forensics Report

Case MT-20261022-005819-5DC9 | Generated 2026-10-21 21:59:06 UTC

1. Executive Summary

No Significant Findings | Risk Score 13/100

CRITICAL	HIGH	MEDIUM	LOW	INFO
0	0	0	8	1

MemTrace analyzed the memory image **MemoryDump_Lab1.raw** and identified **9** finding(s) across 48 recovered processes, 118 network artifacts and 11 suspicious memory regions. No critical or high severity indicators were identified by the automated rules.

2. Evidence and Methodology

Evidence Item

File Name	MemoryDump_Lab1.raw
Size	1023.9 MB (1,073,676,288 bytes)
SHA-256	`88e708413898c98351d1a5a221799fc34460dfecb20e482ca5247b0b113b8cf2`
MD5	b9fec1a443907d870cb32b048bda9380
Integrity	Verified - hash unchanged after analysis

Analysis Environment

Tool	MemTrace 1.0.0
Analysis Engine	Volatility 2.28.2
Python	3.13.1
Analyst OS	Windows 11
Started (UTC)	2026-10-21T21:58:19+00:00
Finished (UTC)	2026-10-21T21:59:06+00:00

Target System Profile

64-bit	True
PAE	False
NT Major Version	6
NT Minor Version	1
Product Type	NtProductWinNt
System Root	C:\Windows
System Time at Capture	2019-12-11 14:38:00+00:00
Build Lab	7601.17514.amd64fre.win7sp1_rtm.
Processors	1

Methodology

The evidence was hashed before and after analysis to demonstrate that it was not modified. Volatility 3 plugins were executed against the image to enumerate processes from the active process list (pslist) and by pool scanning (psscan), reconstruct the process tree and command lines, list loaded modules (dlllist), read the AppInit_DLLs registry value from the in-memory registry (printkey), recover network artifacts (netscan) and locate private executable memory regions (malfind). MemTrace detection rules then compared these sources to identify hidden processes, masquerading processes and modules, abnormal parent-child relationships, DLL injection persistence, suspicious network activity and code injection.

Plugin	Status	Detail	Time (s)
windows.info	OK	23 rows	1.3
windows.pslist	OK	48 rows	1.4
windows.psscan	OK	56 rows	5.8
windows.pstree	OK	48 rows	1.2
windows.cmdline	OK	48 rows	1.2
windows.dlllist	OK	2305 rows	3.5
windows.netscan	OK	118 rows	6.6
windows.malware.malfind	OK	11 rows	16.5
windows.registry.printkey --key "Microsoft\Windows NT\CurrentVersion\Windows"	OK	27 rows	2.3
windows.registry.printkey --key "Wow6432Node\Microsoft\Windows NT\CurrentVersion\Windows"	OK	27 rows	2.1

3. Detailed Findings

LOW F-001 Low-confidence executable memory in svchost.exe (PID 472)

Executable regions were flagged by malfind but contain empty memory, a Windows heap header, sparse data, the x64 call-forwarding thunks that Windows creates for RPC and COM, or code generated by a known just-in-time compiler (browsers, .NET, Java, antimalware engines). These are common on clean systems and rarely indicate injection on their own.

Category	Injection
Rule	RWX_MEMORY_LOW_CONFIDENCE
Process	svchost.exe (PID 472)
MITRE ATT&CK	T1055
Regions	1
Empty regions	0
Heap regions	0
Thunk regions	1
Data regions	0
JIT code regions	0
Addresses	0xb30000

Recommendation: No action needed unless other findings involve this process.

LOW F-002 Low-confidence executable memory in explorer.exe (PID 604)

Executable regions were flagged by malfind but contain empty memory, a Windows heap header, sparse data, the x64 call-forwarding thunks that Windows creates for RPC and COM, or code generated by a known just-in-time compiler (browsers, .NET, Java, antimalware engines). These are common on clean systems and rarely indicate injection on their own.

Category	Injection
Rule	RWX_MEMORY_LOW_CONFIDENCE
Process	explorer.exe (PID 604)
MITRE ATT&CK	T1055
Regions	2
Empty regions	1
Heap regions	0
Thunk regions	1
Data regions	0
JIT code regions	0
Addresses	0x2810000, 0x3db0000

Recommendation: No action needed unless other findings involve this process.

LOW F-003 Low-confidence executable memory in svchost.exe (PID 816)

Executable regions were flagged by malfind but contain empty memory, a Windows heap header, sparse data, the x64 call-forwarding thunks that Windows creates for RPC and COM, or code generated by a known just-in-time compiler (browsers, .NET, Java, antimalware engines). These are common on clean systems and rarely indicate injection on their own.

Category	Injection
Rule	RWX_MEMORY_LOW_CONFIDENCE
Process	svchost.exe (PID 816)
MITRE ATT&CK	T1055
Regions	1
Empty regions	0
Heap regions	0
Thunk regions	1
Data regions	0
JIT code regions	0
Addresses	0xdf0000

Recommendation: No action needed unless other findings involve this process.

LOW F-004 Low-confidence executable memory in svchost.exe (PID 948)

Executable regions were flagged by malfind but contain empty memory, a Windows heap header, sparse data, the x64 call-forwarding thunks that Windows creates for RPC and COM, or code generated by a known just-in-time compiler (browsers, .NET, Java, antimalware engines). These are common on clean systems and rarely indicate injection on their own.

Category	Injection
Rule	RWX_MEMORY_LOW_CONFIDENCE
Process	svchost.exe (PID 948)
MITRE ATT&CK	T1055
Regions	2
Empty regions	0
Heap regions	0
Thunk regions	0
Data regions	1
JIT code regions	1
Addresses	0x23f0000, 0x4c90000

Recommendation: No action needed unless other findings involve this process.

LOW F-005 Low-confidence executable memory in SearchFilterHo (PID 1720)

Executable regions were flagged by malfind but contain empty memory, a Windows heap header, sparse data, the x64 call-forwarding thunks that Windows creates for RPC and COM, or code generated by a known just-in-time compiler (browsers, .NET, Java, antimalware engines). These are common on clean systems and rarely indicate injection on their own.

Category	Injection
Rule	RWX_MEMORY_LOW_CONFIDENCE
Process	SearchFilterHo (PID 1720)
MITRE ATT&CK	T1055
Regions	1
Empty regions	0
Heap regions	1
Thunk regions	0
Data regions	0
JIT code regions	0
Addresses	0x9b0000

Recommendation: No action needed unless other findings involve this process.

LOW F-006 Low-confidence executable memory in wmpnetwk.exe (PID 1856)

Executable regions were flagged by malfind but contain empty memory, a Windows heap header, sparse data, the x64 call-forwarding thunks that Windows creates for RPC and COM, or code generated by a known just-in-time compiler (browsers, .NET, Java, antimalware engines). These are common on clean systems and rarely indicate injection on their own.

Category	Injection
Rule	RWX_MEMORY_LOW_CONFIDENCE
Process	wmpnetwk.exe (PID 1856)
MITRE ATT&CK	T1055
Regions	1
Empty regions	0
Heap regions	0
Thunk regions	1
Data regions	0
JIT code regions	0
Addresses	0xb80000

Recommendation: No action needed unless other findings involve this process.

LOW F-007 Low-confidence executable memory in mspaint.exe (PID 2424)

Executable regions were flagged by malfind but contain empty memory, a Windows heap header, sparse data, the x64 call-forwarding thunks that Windows creates for RPC and COM, or code generated by a known just-in-time compiler (browsers, .NET, Java, antimalware engines). These are common on clean systems and rarely indicate injection on their own.

Category	Injection
Rule	RWX_MEMORY_LOW_CONFIDENCE
Process	mspaint.exe (PID 2424)
MITRE ATT&CK	T1055
Regions	1
Empty regions	1
Heap regions	0
Thunk regions	0
Data regions	0
JIT code regions	0
Addresses	0x2150000

Recommendation: No action needed unless other findings involve this process.

LOW F-008 Low-confidence executable memory in explorer.exe (PID 2504)

Executable regions were flagged by malfind but contain empty memory, a Windows heap header, sparse data, the x64 call-forwarding thunks that Windows creates for RPC and COM, or code generated by a known just-in-time compiler (browsers, .NET, Java, antimalware engines). These are common on clean systems and rarely indicate injection on their own.

Category	Injection
Rule	RWX_MEMORY_LOW_CONFIDENCE
Process	explorer.exe (PID 2504)
MITRE ATT&CK	T1055
Regions	2
Empty regions	1
Heap regions	0
Thunk regions	1
Data regions	0
JIT code regions	0
Addresses	0x3ec0000, 0x3eb0000

Recommendation: No action needed unless other findings involve this process.

INFO

F-009 Memory acquisition tool running: DumpIt.exe

A known memory acquisition tool was running when the image was captured. This is normally the tool the responder used to create this memory image and is expected.

Category	Process
Rule	ACQUISITION_TOOL
Process	DumpIt.exe (PID 796)
Path	`C:\Users\SmartNet\Downloads\DumpIt\DumpIt.exe`
Create Time	2019-12-11T14:37:54+00:00

Recommendation: Confirm with the responder that this tool was used for the acquisition.

4. Process Listing

PID	PPID	Name	Created	Source	Flags
4	0	System	2019-12-11T13:41:25	pslist	
248	4	smss.exe	2019-12-11T13:41:25	pslist	
296	484	taskhost.exe	2019-12-11T14:32:24	pslist	
320	312	csrss.exe	2019-12-11T13:41:32	pslist	
368	360	csrss.exe	2019-12-11T13:41:33	pslist	
376	248	psxss.exe	2019-12-11T13:41:33	pslist	
416	360	winlogon.exe	2019-12-11T13:41:34	pslist	
424	312	wininit.exe	2019-12-11T13:41:34	pslist	
472	484	svchost.exe	2019-12-11T13:41:47	pslist	
480	484	SearchIndexer.	2019-12-11T14:16:09	pslist	
484	424	services.exe	2019-12-11T13:41:35	pslist	
492	424	lsass.exe	2019-12-11T13:41:35	pslist	
500	424	lsm.exe	2019-12-11T13:41:35	pslist	
588	484	svchost.exe	2019-12-11T13:41:39	pslist	
604	2016	explorer.exe	2019-12-11T14:32:25	pslist	
652	484	VBoxService.ex	2019-12-11T13:41:40	pslist	
720	484	svchost.exe	2019-12-11T13:41:41	pslist	
796	604	DumpIt.exe	2019-12-11T14:37:54	pslist	acquisition-tool
816	484	svchost.exe	2019-12-11T13:41:42	pslist	
852	484	svchost.exe	2019-12-11T13:41:43	pslist	
876	484	svchost.exe	2019-12-11T13:41:43	pslist	
948	484	svchost.exe	2019-12-11T14:16:07	pslist	
1044	484	svchost.exe	2019-12-11T13:41:48	pslist	
1208	484	spoolsv.exe	2019-12-11T13:41:51	pslist	
1248	484	svchost.exe	2019-12-11T13:41:52	pslist	
1372	484	svchost.exe	2019-12-11T13:41:54	pslist	
1416	484	TCPSVCS.EXE	2019-12-11T13:41:55	pslist	
1508	484	sppsvc.exe	2019-12-11T14:16:06	pslist	
1512	2504	WinRAR.exe	2019-12-11T14:37:23	pslist	
1720	480	SearchFilterHo	2019-12-11T14:37:21	pslist	
1844	604	VBoxTray.exe	2019-12-11T14:32:35	pslist	
1856	484	wmpnetwk.exe	2019-12-11T14:16:08	pslist	
1984	604	cmd.exe	2019-12-11T14:34:54	pslist	
1988	852	dwm.exe	2019-12-11T14:32:25	pslist	
2064	816	audiodg.exe	2019-12-11T14:32:37	pslist	
2260	368	conhost.exe	2019-12-11T14:37:54	pslist	
2304	2504	VBoxTray.exe	2019-12-11T14:37:14	pslist	
2368	484	svchost.exe	2019-12-11T14:32:51	pslist	
2424	604	mspaint.exe	2019-12-11T14:35:14	pslist	
2504	3000	explorer.exe	2019-12-11T14:37:14	pslist	
2524	480	SearchProtocol	2019-12-11T14:37:21	pslist	
2660	484	svchost.exe	2019-12-11T14:35:14	pslist	

PID	PPID	Name	Created	Source	Flags
2692	368	conhost.exe	2019-12-11T14:34:54	pslist	
2760	2680	csrss.exe	2019-12-11T14:37:05	pslist	
2808	2680	winlogon.exe	2019-12-11T14:37:05	pslist	
2868	480	SearchProtocol	2019-12-11T14:37:23	pslist	
2908	484	taskhost.exe	2019-12-11T14:37:13	pslist	
3004	852	dwm.exe	2019-12-11T14:37:14	pslist	

5. Network Artifacts

Proto	Local	Remote	State	PID	Owner
UDPv4	0.0.0.0:3702	*:0		472	svchost.exe
UDPv6	:::3702	*:0		472	svchost.exe
UDPv4	0.0.0.0:3702	*:0		472	svchost.exe
UDPv6	:::3702	*:0		472	svchost.exe
UDPv4	0.0.0.0:0	*:0		2368	svchost.exe
UDPv6	:::0	*:0		2368	svchost.exe
UDPv4	0.0.0.0:0	*:0		2368	svchost.exe
UDPv6	:::0	*:0		2368	svchost.exe
UDPv4	0.0.0.0:0	*:0		2368	svchost.exe
UDPv6	:::0	*:0		2368	svchost.exe
UDPv4	0.0.0.0:59438	*:0		472	svchost.exe
UDPv6	:::59438	*:0		472	svchost.exe
UDPv4	0.0.0.0:59437	*:0		472	svchost.exe
TCPv4	0.0.0.0:19	0.0.0.0:0	LISTENING	1416	TCPSVCS.EXE
UDPv4	0.0.0.0:19	*:0		1416	TCPSVCS.EXE
UDPv6	:::19	*:0		1416	TCPSVCS.EXE
UDPv4	0.0.0.0:19	*:0		1416	TCPSVCS.EXE
TCPv4	0.0.0.0:19	0.0.0.0:0	LISTENING	1416	TCPSVCS.EXE
TCPv6	:::19	:::0	LISTENING	1416	TCPSVCS.EXE
TCPv4	0.0.0.0:17	0.0.0.0:0	LISTENING	1416	TCPSVCS.EXE
UDPv4	0.0.0.0:17	*:0		1416	TCPSVCS.EXE
UDPv6	:::17	*:0		1416	TCPSVCS.EXE
UDPv4	0.0.0.0:17	*:0		1416	TCPSVCS.EXE
TCPv4	0.0.0.0:17	0.0.0.0:0	LISTENING	1416	TCPSVCS.EXE
TCPv6	:::17	:::0	LISTENING	1416	TCPSVCS.EXE
UDPv4	0.0.0.0:49194	*:0		1372	svchost.exe
UDPv4	0.0.0.0:49195	*:0		1372	svchost.exe
UDPv6	:::49195	*:0		1372	svchost.exe
TCPv4	0.0.0.0:49156	0.0.0.0:0	LISTENING	484	services.exe
TCPv6	:::49156	:::0	LISTENING	484	services.exe
TCPv4	0.0.0.0:49156	0.0.0.0:0	LISTENING	484	services.exe
TCPv4	0.0.0.0:445	0.0.0.0:0	LISTENING	4	System
TCPv6	:::445	:::0	LISTENING	4	System
UDPv4	0.0.0.0:0	*:0		1044	svchost.exe

Proto	Local	Remote	State	PID	Owner
UDPv6	:::0	*:0		1044	svchost.exe
UDPv4	0.0.0.0:5355	*:0		1044	svchost.exe
UDPv6	:::5355	*:0		1044	svchost.exe
UDPv4	10.0.2.15:137	*:0		4	System
TCPv4	0.0.0.0:3587	0.0.0.0:0	LISTENING	2368	svchost.exe
TCPv6	:::3587	:::0	LISTENING	2368	svchost.exe
UDPv4	0.0.0.0:3702	*:0		472	svchost.exe
UDPv4	0.0.0.0:59435	*:0		472	svchost.exe
UDPv4	0.0.0.0:3702	*:0		472	svchost.exe
UDPv4	0.0.0.0:59436	*:0		472	svchost.exe
UDPv6	:::59436	*:0		472	svchost.exe
UDPv4	0.0.0.0:5004	*:0		1856	wmpnetwk.exe
UDPv4	0.0.0.0:5004	*:0		1856	wmpnetwk.exe
UDPv6	:::5004	*:0		1856	wmpnetwk.exe
UDPv4	0.0.0.0:5005	*:0		1856	wmpnetwk.exe
UDPv6	:::5005	*:0		1856	wmpnetwk.exe
UDPv4	0.0.0.0:3702	*:0		1372	svchost.exe
UDPv6	:::3702	*:0		1372	svchost.exe
UDPv4	10.0.2.15:138	*:0		4	System
UDPv4	127.0.0.1:64388	*:0		1044	svchost.exe
UDPv4	0.0.0.0:5355	*:0		1044	svchost.exe
TCPv4	0.0.0.0:49154	0.0.0.0:0	LISTENING	492	lsass.exe
TCPv4	0.0.0.0:49154	0.0.0.0:0	LISTENING	492	lsass.exe
TCPv6	:::49154	:::0	LISTENING	492	lsass.exe
TCPv4	0.0.0.0:49155	0.0.0.0:0	LISTENING	876	svchost.exe
TCPv6	:::49155	:::0	LISTENING	876	svchost.exe
TCPv4	0.0.0.0:49155	0.0.0.0:0	LISTENING	876	svchost.exe
TCPv4	0.0.0.0:554	0.0.0.0:0	LISTENING	1856	wmpnetwk.exe
TCPv6	:::554	:::0	LISTENING	1856	wmpnetwk.exe
TCPv4	0.0.0.0:554	0.0.0.0:0	LISTENING	1856	wmpnetwk.exe
TCPv4	0.0.0.0:2869	0.0.0.0:0	LISTENING	4	System
TCPv6	:::2869	:::0	LISTENING	4	System
UDPv4	0.0.0.0:3540	*:0		2368	svchost.exe
UDPv6	:::3540	*:0		2368	svchost.exe
TCPv6	::1:49163	::1:2869	CLOSED	1856	wmpnetwk.exe
UDPv4	0.0.0.0:3702	*:0		1372	svchost.exe
UDPv4	0.0.0.0:0	*:0		1416	TCPSVCS.EXE
UDPv4	0.0.0.0:3702	*:0		1372	svchost.exe
TCPv4	0.0.0.0:5357	0.0.0.0:0	LISTENING	4	System
TCPv6	:::5357	:::0	LISTENING	4	System
TCPv4	0.0.0.0:9	0.0.0.0:0	LISTENING	1416	TCPSVCS.EXE
TCPv6	:::9	:::0	LISTENING	1416	TCPSVCS.EXE
UDPv4	0.0.0.0:9	*:0		1416	TCPSVCS.EXE
TCPv4	0.0.0.0:9	0.0.0.0:0	LISTENING	1416	TCPSVCS.EXE

Proto	Local	Remote	State	PID	Owner
TCPv4	0.0.0.0:13	0.0.0.0:0	LISTENING	1416	TCPSVCS.EXE
UDPv4	0.0.0.0:9	*:0		1416	TCPSVCS.EXE
UDPv6	:::9	*:0		1416	TCPSVCS.EXE
TCPv4	0.0.0.0:7	0.0.0.0:0	LISTENING	1416	TCPSVCS.EXE
UDPv4	0.0.0.0:7	*:0		1416	TCPSVCS.EXE
TCPv4	0.0.0.0:7	0.0.0.0:0	LISTENING	1416	TCPSVCS.EXE
TCPv6	:::7	:::0	LISTENING	1416	TCPSVCS.EXE
UDPv4	0.0.0.0:7	*:0		1416	TCPSVCS.EXE
UDPv6	:::7	*:0		1416	TCPSVCS.EXE
UDPv4	0.0.0.0:13	*:0		1416	TCPSVCS.EXE
UDPv6	:::13	*:0		1416	TCPSVCS.EXE
UDPv4	0.0.0.0:13	*:0		1416	TCPSVCS.EXE
TCPv4	0.0.0.0:13	0.0.0.0:0	LISTENING	1416	TCPSVCS.EXE
TCPv6	:::13	:::0	LISTENING	1416	TCPSVCS.EXE
UDPv6	::1:59432	*:0		1372	svchost.exe
UDPv6	fe80::b137:133f:8d0b:8cfe:59431	*:0		1372	svchost.exe
TCPv4	0.0.0.0:10243	0.0.0.0:0	LISTENING	4	System
TCPv6	:::10243	:::0	LISTENING	4	System
UDPv4	10.0.2.15:1900	*:0		1372	svchost.exe
UDPv6	::1:1900	*:0		1372	svchost.exe
UDPv4	127.0.0.1:1900	*:0		1372	svchost.exe
TCPv4	10.0.2.15:139	0.0.0.0:0	LISTENING	4	System
TCPv4	0.0.0.0:135	0.0.0.0:0	LISTENING	720	svchost.exe
TCPv6	:::135	:::0	LISTENING	720	svchost.exe
TCPv4	0.0.0.0:49152	0.0.0.0:0	LISTENING	424	wininit.exe
TCPv4	0.0.0.0:135	0.0.0.0:0	LISTENING	720	svchost.exe
TCPv4	0.0.0.0:49152	0.0.0.0:0	LISTENING	424	wininit.exe
TCPv6	:::49152	:::0	LISTENING	424	wininit.exe
UDPv4	0.0.0.0:3702	*:0		1372	svchost.exe
UDPv6	:::3702	*:0		1372	svchost.exe
TCPv6	::1:2869	::1:49163	CLOSED	4	System
TCPv4	0.0.0.0:49153	0.0.0.0:0	LISTENING	816	svchost.exe
TCPv4	0.0.0.0:49153	0.0.0.0:0	LISTENING	816	svchost.exe
TCPv6	:::49153	:::0	LISTENING	816	svchost.exe
UDPv4	0.0.0.0:5005	*:0		1856	wmpnetwk.exe
UDPv4	10.0.2.15:59433	*:0		1372	svchost.exe
UDPv6	fe80::b137:133f:8d0b:8cfe:1900	*:0		1372	svchost.exe
UDPv4	127.0.0.1:59434	*:0		1372	svchost.exe
TCPv6	:0	4800:ca00:80fa:ffff:4800:ca00:80fa:ffff:0	CLOSED	1044	svchost.exe
UDPv4	0.0.0.0:0	*:0		652	VBoxService.ex

6. Limitations

This report was produced by automated analysis. Findings are investigative leads that should be validated by a qualified examiner. Absence of findings does not prove the system was not compromised; memory captures only a point-in-time state and advanced malware may evade these detection rules.